📘 FLIPKART CUSTOMER SEGMENTATION ANALYSIS

Project Documentation

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1. Executive Summary

The **Flipkart Customer Segmentation Analysis** project focuses on understanding customer purchasing behavior, identifying valuable customer groups, and generating actionable business insights through data analytics and interactive dashboarding.

Customer segmentation helps businesses categorize customers based on common characteristics such as purchasing frequency, spending patterns, product preferences, and payment behavior. By identifying these segments, businesses can implement personalized marketing strategies, improve customer retention, and increase overall profitability.

This project utilizes Flipkart sales transaction data and Power BI visualizations to provide a comprehensive view of customer behavior and business performance.

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2. Project Title

Flipkart Customer Segmentation Analysis using Power BI and Python

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3. Problem Statement

E-commerce platforms like Flipkart serve thousands of customers with varying shopping behaviors and preferences.

Without customer segmentation, businesses face challenges such as:

- Inefficient marketing campaigns
- Poor customer retention
- Low customer engagement
- Generic promotional strategies
- Difficulty identifying high-value customers

The objective of this project is to analyze customer transaction data and segment customers based on their purchasing behavior to support strategic business decision-making.

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4. Business Objectives

The primary objectives of this project are:

Customer Analysis

- Understand customer purchasing behavior.
- Identify valuable customers.

Revenue Analysis

- Analyze revenue contribution across customers, cities, and categories.

Product Analysis

- Identify top-performing products and categories.

Payment Analysis

- Understand customer payment preferences.

Customer Segmentation

- Group customers into High, Medium, and Low-value segments.

Business Recommendations

- Provide data-driven recommendations for improving revenue and customer retention.

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5. Dataset Overview

The dataset consists of Flipkart sales transactions and customer purchase records.

Dataset Attributes

Column Name	Description
Customer_ID	Unique customer identifier
Customer_Name	Customer name
Age	Customer age
Gender	Customer gender
City	Customer location
Order_ID	Unique order identifier
Product_Name	Purchased product
Category	Product category
Quantity	Number of items purchased
Revenue	Revenue generated
Payment_Method	Payment mode
Rating	Product rating
Order_Date	Purchase date

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6. Data Cleaning & Preparation

The dataset was cleaned before analysis to ensure accuracy and reliability.

Data Cleaning Steps

Missing Value Handling

- Checked for null values.
- Removed or corrected incomplete records.

Duplicate Removal

- Identified duplicate transactions.
- Removed duplicate entries.

Data Type Validation

- Converted Order Date into Date format.
- Verified numerical fields.

Revenue Validation

- Checked for negative and abnormal values.

Standardization

- Standardized category names.
- Standardized city names.

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7. Methodology

The project follows a structured analytics workflow:

```
Data Collection
    ↓
Data Cleaning
    ↓
Exploratory Data Analysis
    ↓
Customer Segmentation
    ↓
Dashboard Development
    ↓
Business Insights
    ↓
Recommendations
```

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8. Customer Segmentation Logic

Customers were classified based on purchasing frequency and revenue contribution.

High-Value Customers

Characteristics:

- Frequent purchases
- High spending
- Significant revenue contribution

Business Importance:

- Most profitable customers
- Priority for retention campaigns

Medium-Value Customers

Characteristics:

- Moderate purchase frequency
- Moderate spending behavior

Business Importance:

- Potential future high-value customers

Low-Value Customers

Characteristics:

- Low purchase frequency
- Low revenue contribution

Business Importance:

- Require re-engagement campaigns

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9. Dashboard Overview

The Power BI dashboard provides a complete overview of customer behavior and business performance.

Page Components

KPI Section

Contains:

- Total Revenue
- Total Orders
- Total Customers
- Average Rating

Purpose:

Provide a quick business performance summary.

Revenue by City

Shows revenue contribution from:

- Mumbai
- Chennai
- Hyderabad
- Delhi
- Kolkata
- Bengaluru

Purpose:

Identify top-performing cities.

Revenue by Month

Displays monthly revenue trends.

Purpose:

Analyze seasonality and demand fluctuations.

Revenue by Category

Displays category contribution:

- Home
- Fashion
- Beauty
- Electronics

Purpose:

Understand category performance.

Payment Method Analysis

Displays order distribution by:

- UPI
- Credit Card
- Debit Card
- COD
- EMI

Purpose:

Analyze customer payment preferences.

Customer Segmentation Chart

Displays:

- High Segment Customers
- Medium Segment Customers
- Low Segment Customers

Purpose:

Understand customer distribution.

Top Products Analysis

Displays:

- Product Name
- Quantity Sold
- Revenue Generated

Purpose:

Identify best-selling products.

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10. DAX Measures Used

Basic Measures

Total Revenue

```
Total Revenue =
SUM('Flipkart_Clean_Dataset'[Revenue])
```

Total Orders

Total Orders =
COUNT('Flipkart_Clean_Dataset'[Order_ID])

Total Customers

Total Customers =
DISTINCTCOUNT('Flipkart_Clean_Dataset'[Customer_ID])

Average Rating

Average Rating =
AVERAGE('Flipkart_Clean_Dataset'[Rating])

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11. Additional DAX Measures

Revenue Per Customer

Revenue Per Customer =
DIVIDE([Total Revenue],[Total Customers])

Average Order Value

Average Order Value =
DIVIDE([Total Revenue],[Total Orders])

Total Quantity Sold

Total Quantity Sold =
SUM('Flipkart_Clean_Dataset'[Quantity])

Orders Per Customer

Orders Per Customer =
DIVIDE([Total Orders],[Total Customers])

Revenue Growth %

```
Revenue Growth % =
VAR PrevMonth =
CALCULATE(
    [Total Revenue],
    DATEADD('Calendar'[Date],-1,MONTH)
)
RETURN
DIVIDE([Total Revenue]-PrevMonth,PrevMonth)
```

Average Revenue Per Order

Average Revenue Per Order =
DIVIDE([Total Revenue],[Total Orders])

Customer Retention Index

Customer Retention Index =
DIVIDE([Total Orders],[Total Customers])

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12. Key Findings

Revenue Analysis

- Total revenue exceeded ₹4 Million.
- Revenue concentration is higher in metropolitan cities.
- Mumbai contributes the highest revenue.

Customer Analysis

- High-value customers generate a major share of revenue.
- Medium-value customers represent the largest customer group.
- Low-value customers require engagement strategies.

Product Analysis

- Home category contributes the highest revenue.
- Electronics category performs consistently well.
- Certain products significantly outperform others.

Payment Analysis

- UPI is one of the most preferred payment methods.
- Card-based payments show strong adoption.
- EMI transactions represent a smaller share.

Customer Satisfaction

- Average rating is above 4.
- Customer satisfaction levels are generally positive.

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13. Business Insights

Insight 1

A small percentage of customers contribute a large percentage of revenue.

Insight 2

Major cities dominate overall sales performance.

Insight 3

Home and Electronics categories are revenue drivers.

Insight 4

Digital payment methods are becoming customer preferences.

Insight 5

Monthly revenue fluctuations indicate seasonal demand patterns.

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14. Recommendations

Recommendation 1: Loyalty Program

Develop premium loyalty programs for high-value customers.

Benefits:

- Improved retention
- Increased customer lifetime value

Recommendation 2: Upselling Strategy

Target medium-value customers with:

- Product bundles
- Personalized offers
- Cross-selling campaigns

Benefits:

- Higher average order value

Recommendation 3: Re-Engagement Campaigns

Target low-value customers using:

- Discount vouchers
- Festival offers
- Email marketing campaigns

Benefits:

- Increased purchase frequency

Recommendation 4: Inventory Optimization

Increase stock availability for top-selling products and categories.

Benefits:

- Reduced stock-outs
- Improved sales performance

Recommendation 5: Digital Payment Promotion

Provide cashback and rewards for UPI and card transactions.

Benefits:

- Faster transactions
- Improved customer experience

Recommendation 6: City-Based Marketing

Allocate higher marketing budgets to top-performing cities while exploring growth opportunities in lower-performing regions.

Benefits:

- Better marketing ROI
- Increased market penetration

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15. Future Scope

The project can be enhanced by implementing:

RFM Analysis

- Recency
- Frequency
- Monetary Value

K-Means Clustering

- Machine Learning-based customer segmentation

Customer Lifetime Value Prediction

- Long-term customer profitability estimation

Churn Prediction

- Identify customers likely to stop purchasing

Recommendation System

- Personalized product recommendations

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16. Conclusion

The **Flipkart Customer Segmentation Analysis** project successfully demonstrates how customer transaction data can be transformed into meaningful business insights using Power BI and analytics techniques.

The dashboard enables stakeholders to:

- Understand customer behavior
- Identify valuable customer groups
- Monitor revenue trends
- Analyze product performance
- Evaluate payment preferences
- Make data-driven business decisions

By leveraging customer segmentation, Flipkart can improve marketing effectiveness, increase customer retention, optimize resource allocation, and drive sustainable business growth.